

DUKE FIELD, FL, USA

WMO: 722246

Lat: 30.650N

Lon: 86.517W

Elev: 190

StdP: 14.60

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 03844

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.1	28.2	9.9	9.2	36.0	14.4	11.5	39.3	23.7	59.8	20.5	53.7	3.7	340	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	95.1	76.2	93.1	76.2	91.3	75.9	80.5	88.5	79.5	87.4	78.5	86.2	7.5	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	150.5	83.3	77.4	143.5	82.5	76.8	140.6	82.0	44.1	88.6	43.1	87.4	42.1	86.3	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 19.0	(b) 16.9	(c) 15.2	(e) 18.7	(f) 98.0	(g) 3.6	(h) 2.3	(i) 16.1	(j) 99.6	(k) 14.0	(l) 100.9	(m) 11.9	(n) 102.2	(o) 9.3	(p) 103.9		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			17.5	82.4	3.8	1.1	14.7	83.2	12.5	83.8	10.4	84.4	7.6	85.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.2	50.0	54.5	60.0	66.2	74.1	80.2	81.3	81.1	78.0	68.5	57.6	54.4
	DBStd	13.17	10.52	9.42	8.45	5.85	5.02	3.22	2.68	2.80	4.13	7.48	8.20	9.67
	HDD50	304	136	59	17	0	0	0	0	0	0	1	24	67
	HDD65	1694	475	309	198	55	7	0	0	0	0	52	249	349
	CDD50	6584	134	184	328	487	746	906	970	963	839	574	251	202
	CDD65	2500	9	14	44	91	288	456	505	498	390	160	26	19
	CDH74	23521	7	43	268	832	2751	4629	5024	4763	3467	1513	181	43
	CDH80	9563	0	1	31	139	1054	2128	2215	2039	1477	464	15	0
(14) Wind	WSAvg	6.0	6.9	7.0	7.4	7.2	6.3	5.4	4.9	4.6	4.7	5.5	5.6	6.5
(15) Precipitation	PrecAvg	59.3	4.3	4.6	5.3	5.1	3.3	5.4	7.0	6.2	5.4	3.7	4.1	4.8
	PrecMax	82.4	16.5	8.3	12.8	16.5	11.6	12.0	15.8	9.8	25.4	12.0	11.9	11.4
	PrecMin	38.4	0.2	1.3	0.6	1.0	0.4	1.4	2.6	1.8	1.4	0.2	0.4	1.9
	PrecStd	12.6	3.1	2.0	3.3	3.2	2.8	2.8	3.2	2.0	4.7	3.4	3.2	2.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.0	78.5	82.9	86.0	94.7	99.0	96.5	97.0	95.8	90.6	82.1	77.3
		MCWB	66.8	66.6	66.3	69.1	72.5	75.4	77.2	76.4	74.4	73.3	68.8	68.8
	2%	DB	72.5	75.0	80.4	83.5	90.8	94.9	94.1	93.7	92.7	87.8	78.4	75.1
		MCWB	66.7	65.2	65.7	67.2	72.0	76.2	77.3	77.5	74.6	71.9	67.3	68.2
	5%	DB	70.3	72.1	77.0	81.4	88.1	92.6	91.8	91.7	90.3	85.3	75.7	72.7
		MCWB	65.5	65.0	64.8	66.2	71.4	75.7	76.9	77.3	75.2	71.6	65.3	68.0
	10%	DB	67.6	69.7	73.4	79.2	86.0	90.3	90.1	89.9	88.2	82.2	73.1	70.1
		MCWB	63.5	65.0	62.9	66.0	71.2	75.6	76.6	77.0	75.3	69.5	64.7	66.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.8	71.0	71.7	74.9	77.4	80.5	81.1	81.5	82.3	78.6	74.4	73.5
		MCDB	72.4	73.2	76.4	79.1	85.6	89.8	90.4	90.4	88.1	83.1	77.2	74.9
	2%	WB	69.1	69.3	70.0	72.9	75.6	79.3	79.8	80.5	79.9	76.9	71.6	71.2
		MCDB	70.8	71.6	74.1	76.9	83.2	88.1	88.8	88.8	85.1	81.8	74.6	73.0
	5%	WB	66.9	67.7	68.6	71.5	74.7	78.2	78.9	79.5	78.0	75.3	68.9	69.3
		MCDB	68.9	70.1	72.7	76.6	82.3	87.0	87.4	87.1	84.4	80.9	73.5	71.4
	10%	WB	63.7	65.8	66.6	69.8	73.6	77.4	78.1	78.5	76.8	73.4	66.5	66.8
		MCDB	65.9	68.9	71.5	75.0	81.3	86.0	86.2	85.6	83.6	79.1	71.6	69.4
(35) Mean Daily Temperature Range	5% DB	MDBR	21.9	22.3	23.3	23.5	22.4	19.6	18.2	17.6	20.0	24.3	26.0	20.8
		MCDBR	19.1	20.7	23.5	25.6	25.8	23.0	20.5	20.8	22.8	24.4	25.7	19.5
	5% WB	MCWBR	14.3	13.2	11.8	11.8	9.9	7.8	7.2	7.5	8.3	11.0	15.7	14.2
		MCWBR	16.1	16.8	17.7	16.6	20.2	19.9	18.1	17.3	17.9	18.0	20.2	16.0
(40) Clear-Sky Solar Irradiance	taub	taub	0.329	0.343	0.365	0.393	0.433	0.465	0.507	0.497	0.435	0.380	0.352	0.337
		taud	2.543	2.505	2.438	2.368	2.287	2.253	2.153	2.176	2.324	2.456	2.475	2.529
	Ebn at Noon	Ebn at Noon	287	292	292	285	273	263	252	253	266	275	275	278
		Edh at Noon	27	31	35	39	43	44	49	47	39	32	29	26
(44) All-Sky Solar Radiation	RadAvg	RadAvg	980	1169	1585	1912	2114	1996	1890	1754	1630	1440	1110	860
	RadStd	RadStd	70	151	125	152	125	164	116	129	102	142	97	87

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+277	+175

Nomenclature: See separate page